

# Omar Mohamed Abdelsalam

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## Summary

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Electronics & Communications Engineering student (Helwan University, Class of 2027) with a strong focus on Digital IC Design, RTL modeling, and verification. Currently advancing expertise in ASIC/FPGA design flows, Verilog, and EDA tools (Questa Sim, Vivado). Backed by a solid foundation in C++/Python programming and embedded systems.

## EDUCATION

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- **B.Sc. in Electronics & Communication Engineering – Helwan University** Cairo, Egypt  
Expected  
2027

## Technical SKILLS

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- **Hardware Description Languages (HDLs) & Scripting:** Verilog, Tcl, C++, Embedded C, MATLAB, System Verilog (Datatypes, Interfaces, Threading, Subroutines), Object-Oriented Programming (OOP).
- **EDA Tools & Software:** Siemens Questa Sim, Questa Formal, Xilinx Vivado, Proteus, Multisim
- **Design & Verification Skills:** RTL Modeling, Testbench Development, FPGA Prototyping, Static Timing Analysis (STA), Clock Domain Crossing (CDC), Linting,
- **Microcontrollers & Hardware:** Basys 3 FPGA, Arduino, STM32, PIC, Atmega32, PCB Design, Basics of Formal Verification, FPGA-based Prototyping, Emulation

## COURSES

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- **DIGITAL DESIGN USING VERILOG AND FPGA FLOW USING VIVADO** | *Eng. Kareem Waseem* | **[February] 2026 – [March] 2026**
    - **RTL Design & Modeling:** Designing combinational/sequential logic, FSMs, and memory architecture using Verilog.
    - **Verification & Automation:** Developing test benches and automating simulation flows using Tcl scripting in Siemens Questa Sim; learning static linting and CDC checks.
    - **FPGA Implementation:** Gaining direct experience with Xilinx Vivado and the Basys 3 board, including synthesis, Static Timing Analysis (STA), and IP core integration.
  - **DIGITAL VERIFICATION USING SYSTEMVERILOG AND UVM** | *Eng. Kareem Waseem* | **[February] 2026 – Present**
    - **Module 1: System Verilog & SVA:** OOP basics, System Verilog datatypes, interfaces, threading, subroutines, and System Verilog Assertions (SVA).
    - **Module 2: UVM Architecture:** Simulation-based verification, UVM structures, components, sequences, TLM, Phasing, and the UVM factory.
    - **Module 3: Verification Planning & Coverage:** Requirements extraction, constrained random stimulus generation, functional coverage modeling, and RTL code coverage analysis.
    - **Module 4: Advanced Methodologies:** Introduction to formal verification, FPGA-based prototyping, and emulators.
  - **Embedded Systems Diploma (230 Hours) – Eng. Mohamed Zaghloul**  
Computer Architecture, C Programming Language, Data Structures & Algorithms, AVR Microcontroller Interfacing, Real-Time Operating Systems (RTOS), Testing, Validation & Software Engineering
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## EXPERIENCE

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- **Sales & Business Development Intern – Schneider Electric | Aug 2025 – Sep 2025**

Gained direct experience in sales channels including ODMs, distributors, and system integrators. Explored EcoStruxure architecture with a focus on Energy Management and IoT solutions. Developed a comprehensive understanding of the product sales lifecycle and customer needs. Collaborated with industry professionals, enhancing both technical and commercial skills.

- **Technical Intern – After-Sales & Spare Parts, EGYBRIT | Jul 2025 – Aug 2025**

Diagnosed and repaired industrial machinery faults. Programmed and configured equipment using diagnostic software. Supported maintenance & troubleshooting in materials handling systems. Gained direct insight into integrating electronics, mechanics, and logistics.

- **Network & Systems Security Intern – Americana | Aug 2024 – Sep 2024**

Monitored and secured enterprise networks (firewalls, routers, switches). Conducted vulnerability assessments and supported patch management.

## SOFT SKILLS

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|---------------------------|---------------------|
| • Leadership              | • Critical Thinking |
| • Effective Communication | • Teamwork          |
| • Problem-Solving         | • Adaptability      |

## LANGUAGES

- Arabic — Native
- English — Fluent